



DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

CCW- OPERATING SYSTEMS MCQs

MODULE 1

Q1. Which of the following is NOT a service provided by an operating system?

- (a) File management
- (b) Memory allocation
- (c) Network routing
- (d) Process scheduling

Q2. Which system call is used to terminate a process?

- (a) exec()
- (b) exit()
- (c) fork()
- (d) wait()

Q3. Which of the following describes a microkernel-based operating system?

- (a) All services are executed in user space.
- (b) Only essential services run in kernel mode.
- (c) All services run in kernel mode for efficiency.
- (d) It uses a large number of device drivers.

Q4. Which structure in operating systems is characterized by minimal functionality, with other features implemented as user-level programs?

- (a) Microkernel
- (b) Monolithic kernel
- (c) Layered approach
- (d) Modular structure

Q5. Which of the following is a key advantage of the layered approach to operating system design?

- (a) Flexibility in hardware management
- (b) Easy to understand and maintain
- (c) Simplicity in implementation
- (d) High execution speed

Q6. In the layered approach to operating systems, what is the role of the topmost layer?

- (a) Hardware abstraction

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- (c) User interface management
- (d) Process scheduling
- (e) Memory management

Q7. What is the role of the system call in operating systems?

- (a) To allow user applications to interact with the hardware
- (b) To allow user applications to request kernel-level services
- (c) To manage system resources directly
- (d) To manage user permissions

Q8. Which of the following describes the purpose of a system boot process?

- (a) Initialization of the hardware
- (b) Loading the operating system into memory
- (c) Starting all user applications
- (d) Establishing network connections

Q9. In a simple structure operating system, how are the different components of the OS organized?

- (a) Each component works independently without interaction.
- (b) Components are tightly coupled and interact directly.
- (c) The OS has separate modules for each component.
- (d) All components are arranged in layers.

Q10. Which of the following is a disadvantage of the monolithic kernel structure?

- (a) Complex and difficult to maintain
- (b) High overhead due to system calls
- (c) Lack of flexibility
- (d) High memory usage

Q11. Which state does a process transition to when it is ready to be executed by the CPU?

- (a) Running
- (b) Ready
- (c) Waiting
- (d) Terminated

Q12. What is stored in the process control block (PCB)?

- (a) Program code and data
- (b) Process state and resources
- (c) Input and output buffers
- (d) Memory maps and virtual addresses

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Q13. What is the main purpose of threads in a process?

- (a) To run the entire program code
- (b) To manage resources independently
- (c) To divide a process into smaller, manageable tasks
- (d) To enhance user interface response times

Q14. Which of the following is NOT a valid process state?

- (a) New
- (b) Ready
- (c) Running
- (d) Suspended

Q15. Which of the following describes the role of a process scheduler?

- (a) Allocates memory to processes
- (b) Decides which process gets the CPU next
- (c) Handles process termination
- (d) Initiates system boot

Q16. What is the purpose of inter-process communication (IPC)?

- (a) To synchronize processes to avoid conflicts
- (b) To allow processes to communicate with each other
- (c) To allocate memory for processes
- (d) To manage CPU scheduling

Q17. Which of the following methods is used for inter-process communication where processes access a shared memory region?

- (a) Message passing
- (b) Semaphore synchronization
- (c) Shared memory
- (d) Socket communication

Q18. Which IPC mechanism involves passing messages between processes?

- (a) Shared memory
- (b) Message passing
- (c) Signal synchronization
- (d) File mapping

Q19. Which of the following actions is performed by the `fork()` system call?

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- (a) Creates a new process
- (b) Terminates the current process
- (c) Waits for a child process to terminate
- (d) Allocates resources to a process

Q20. What does the `wait()` system call do in process management?

- (a) Waits for a process to complete before continuing
- (b) Terminates a process after it finishes execution
- (c) Suspends a process
- (d) Allocates resources to a process

MODULE 2

Q21. Threads of a process share

- (a) global variables but not heap.
- (b) heap but not global variables.
- (c) neither global variables nor heap.
- (d) both heap and global variables.

Q22. Which of the following is/are shared by all the threads in a process?

- I. Program counter
- II. Stack
- III. Address space
- IV. Registers

- (a) I and II only
- (b) III only
- (c) IV only
- (d) III and IV only

Q23. Consider the set of processes with arrival time (in milliseconds), CPU burst time (in milliseconds), and priority (0 is the highest priority) shown below. None of the processes have I/O burst time.

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Process	Arrival Time	Burst Time	Priority
P_1	0	11	2
P_2	5	28	0
P_3	12	2	3
P_4	2	10	1
P_5	9	16	4

The average waiting time (in milliseconds) of all the processes using preemptive priority scheduling algorithm is

- (a) 29
- (b) 30
- (c) 31
- (d) 32

Q24. Consider the following processes, with the arrival time and the length of the CPU burst given in milliseconds. The scheduling algorithm used is preemptive shortest remaining-time first.

Process	Arrival Time	Burst Time
P_1	0	10
P_2	3	6
P_3	7	1
P_4	8	3

The average turn around time of these processes ismilliseconds.

- (a) 8.25
- (b) 8.26
- (c) 8.27
- (d) 8.28

Q25. The maximum number of processes that can be in Ready state for a computer system with n CPUs is

- (a) n
- (b) n^2
- (c) 2^n
- (d) Independent of n

Q26. A scheduling algorithm assigns priority proportional to the waiting time of a process. Every process starts with priority zero (the lowest priority). The scheduler re-evaluates the process priorities every T time units and decides the next process to schedule. Which one of the following is **TRUE** if the processes have no I/O operations and all arrive at time zero?

- (a) This algorithm is equivalent to the first-come-first-serve algorithm.
- (b) This algorithm is equivalent to the round-robin algorithm.

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(c) This algorithm is equivalent to the shortest-job-first algorithm.

(d) This algorithm is equivalent to the shortest-remaining-time-first algorithm.

Q27. A thread is usually defined as a "light weight process" because an operating system (OS) maintains smaller data structures for a thread than for a process. In relation to this, which of the following is **TRUE** ?

(a) On per-thread basis, the OS maintains only CPU register state

(b) The OS does not maintain a separate stack for each thread

(c) On per-thread basis, the OS does not maintain virtual memory state

(d) On per thread basis, the OS maintains only scheduling and accounting

infor- mation

Q28. Consider the following table of arrival time and burst time for three processes P0, P1 and P2.

Process	Arrival Time	Burst Time
P0	0 ms	9 ms
P1	1 ms	4 ms
P2	2 ms	9 ms

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The pre-emptive shortest job first scheduling algorithm is used. Scheduling is carried out only at arrival or completion of processes. What is the average waiting time for the three processes?

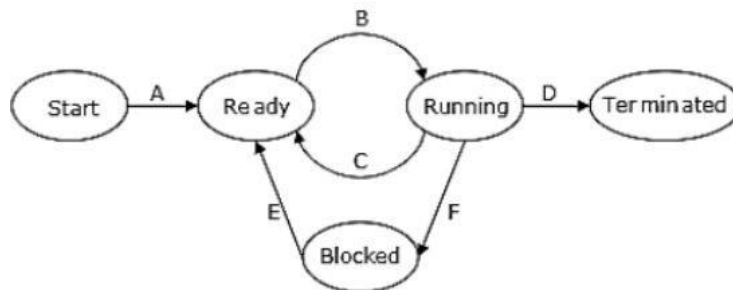
- (a) 5.0 ms
- (b) 4.33 ms
- (c) 6.33 ms
- (d) 7.33 ms

Q29. Which of the following statements are true?

- I. Shortest remaining time first scheduling may cause starvation
- II. Preemptive scheduling may cause starvation
- III. Round robin is better than FCFS in terms of response time

- (a) I only
- (b) I and III only
- (c) II and III only
- (d) I, II and III

Q30. In the following process state transition diagram for a uniprocessor system, assume that there are always some processes in the ready state:



Now consider the following statements:

- I. If a process makes a transition D, it would result in another process making transition A immediately.
- II. A process P2 in blocked state can make transition E while another process P1 is in running state.
- III. The OS uses preemptive scheduling.
- IV. The OS uses non-preemptive scheduling. Which of the above statements are TRUE?

- (a) I and II
- (b) I and III
- (c) II and III
- (d) II and IV

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Q31. Consider the following statements about user level threads and kernel level threads. Which one of the following statements is FALSE?

- (a) Context switch time is longer for kernel level threads than for user level threads.
- (b) User level threads do not need any hardware support.
- (c) Related kernel level threads can be scheduled on different processors in a multi-processor system.
- (d) Blocking one kernel level thread blocks all related threads

Q32. Consider three CPU-intensive processes, which require 10, 20 and 30 time units and arrive at times 0, 2 and 6, respectively. How many context switches are needed if the operating system implements a shortest remaining time first scheduling algorithm? Do not count the context switches at time zero and at the end.

- (a) 1
- (b) 2
- (c) 3
- (d) 4

Q33. Consider the following statements with respect to user-level threads and kernel supported threads

- (i) context switch is faster with kernel-supported threads
 - (ii) for user-level threads, a system call can block the entire process
 - (iii) Kernel supported threads can be scheduled independently
 - (iv) User level threads are transparent to the kernel
- Which of the above statements are true?

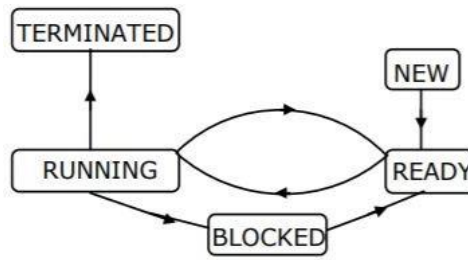
- (a) (ii), (iii) and (iv) only
- (b) (ii) and (iii) only
- (c) (i) and (iii) only
- (d) (i) and (ii) only

Q34. Which of the following scheduling algorithms is non-preemptive?

- (a) Round Robin
- (b) First-In First-Out
- (c) Multilevel Queue Scheduling
- (d) Multilevel Queue Scheduling with Feedback

Q35. The process state transition diagram in below figure is representative of

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- (a) a batch operating system
- (b) an operating system with a preemptive scheduler
- (c) an operating system with a non-preemptive scheduler
- (d) a uni-programmed operating system

Q36. Consider the following statements about process state transitions for a system using preemptive scheduling.

- I. A running process can move to ready state.
- II. A ready process can move to running state.
- III. A blocked process can move to running state.
- IV. A blocked process can move to ready state. Which of the above statements are TRUE?

- (a) II and III only
- (b) I, II and III only
- (c) I, II, III and IV
- (d) I, II and IV only

Q37. The term 'aging' refers to

- (a) booting up the priority of the process in multi-level of queue without feedback
- (b) keeping track of the following a page has been in memory for the purpose of LRU replacement
- (c) letting job reside in memory for a certain amount of time so that the number of pages required can be estimated accurately
- (d) gradually increasing the priority of jobs that wait in the system for a long time to remedy infinite blocking
- (e) In Operating systems, aging is a scheduling technique used to avoid starvation. In this, the priority of the jobs that have a longer waiting time is increased as compared to the newer processes, to avoid the starvation of older processes

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Q38.Round Robin schedule is essentially the preemptive version of

- (a) FIFO
- (b) Shortest job first
- (c) Shortest remaining time
- (d) Longest remaining time

Q39.With Round-Robin CPU scheduling in a time shared system

- (a) using very large time slices (quantas) degenerates into First Come First served (FCFS) algorithm.
- (b) using extremely small time slices improves performance
- (c) using very small time slices degenerates into Last-In First-Out (LIFO) algorithm.
- (d) using medium sized times slices leads to shortest Request time First (SRTF) algorithm

Q40.The performance of Round Robin algorithm depends heavily on

- (a)size of the process
- (b) the I/O bursts of the process
- (c) the CPU bursts of the process
- (d) the size of the time quantum

MODULE 3

Q41. Consider the methods used by processes P1 and P2 for accessing their critical sections whenever needed, as given below. The initial values of shared boolean variables S1 and S2 are randomly assigned.

Method Used by P1	Method Used by P2
while (S1 == S2) ; Critical Section S1 = S2;	while (S1 != S2) ; Critical Section S2 = not (S1);

Which one of the following statements describes the properties achieved?

- (a) Mutual exclusion but not progress
- (b) Progress but not mutual exclusion
- (c) Neither mutual exclusion nor progress
- (d) Both mutual exclusion and progress

Q42. The following program consists of 3 concurrent processes and 3 binary semaphores.The semaphores are initialized as $S_0 = 1$, $S_1 = 0$, $S_2 = 0$.

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Process P0	Process P1	Process P2
<pre>while (true) { wait (S0); print '0' release (S1); release (S2); }</pre>	<pre>wait (S1); Release (S0);</pre>	<pre>wait (S2); release (S0);</pre>

How many times will process P0 print '0'?

- (a) At least twice
- (b) Exactly twice
- (c) Exactly thrice
- (d) Exactly once

Q43. Which of the following is NOT true of deadlock prevention and deadlock avoidance schemes?

- (a) In deadlock prevention, the request for resources is always granted if the resulting state is safe
- (b) In deadlock avoidance, the request for resources is always granted if the result state is safe
- (c) Deadlock avoidance is less restrictive than deadlock prevention
- (d) Deadlock avoidance requires knowledge of resource requirements a priori

Q44. Which of the following is NOT a valid deadlock prevention scheme?

- (a) Release all resources before requesting a new resource
- (b) Number the resources uniquely and never request a lower numbered resource than the last one requested
- (c) Never request a resource after releasing any resource
- (d) Request and all required resources be allocated before execution

Q45. A counting semaphore was initialized to 10. Then 6P (wait) operations and 4V (signal) operations were completed on this semaphore. The resulting value of the semaphore is

- (a) 0
- (b) 8
- (c) 10
- (d) 12

Q46. A computer has six tape drives, with n processes competing for them. Each process may need two drives. What is the maximum value of n for the system to be deadlock free?

- (a) 6
- (b) 5

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(c) 4

(d) 3

Q47. Consider a system having m resources of the same type. These resources are shared by 3 processes A, B and C, which have peak demands of 3, 4 and 6 respectively. For what value of m deadlock will not occur?

(a) 7

(b) 9

(c) 13, 15

(d) 13

(e) 15

Q48. A critical section is a program segment

(a) which should run in a certain specified amount of time

(b) which avoids deadlocks

(c) where shared resources are accessed

(d) which must be enclosed by a pair of semaphore operations, P and V

Q49. At a particular time of computation the value of a counting semaphore is 7. Then 20 P operations and 15 V operations were completed on this semaphore. The resulting value of the semaphore is:

(a) 42

(b) 2

(c) 7

(d) 12

Q50. A computer system has 6 tape drives, with n process completing for them. Each process may need 3 tape drives. The maximum value of n for which the system is guaranteed to be deadlock free is:

(a) 2

(b) 3

(c) 4

(d) 1

Q51. A critical region is

(a) One which is enclosed by a pair of P and V operations on semaphores.

(b) A program segment that has not been proved bug-free.

(c) A program segment that often causes unexpected system crashes.

(d) A program segment where shared resources are accessed.

Q52. An operating system implements a policy that requires a process to release all resources before making a request for another resource. Select the TRUE statement from the following:

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- (a) Both starvation and deadlock can occur
- (b) Starvation can occur but deadlock cannot occur
- (c) Starvation cannot occur but deadlock can occur
- (d) Neither starvation nor deadlock can occur

Q53. Consider a system having 'm' resources of the same type. The resources are shared by 3 processes A, B, C, which have peak time demands of 3, 4, 6 respectively. The minimum value of 'm' that ensures that deadlock will never occur is

- (a) 11
- (b) 12
- (c) 13
- (d) 14

Q54. Semaphores

- (a) synchronize critical resources to prevent deadlock
- (b) synchronize critical resources to prevent contention
- (c) are used to do I/O
- (d) are used for memory management

Q55. Mutual exclusion problem occurs

- (a) Between two disjoint processes that do not interact
- (b) Among processes that share resources
- (c) Among processes that do not use the same resource
- (d) Between two processes that use different resources of different machine

Q56. In which of the following four necessary conditions for deadlock processes claim exclusive control of the resources they require?

- (a) no preemption
- (b) mutual exclusion
- (c) circular wait
- (d) hold and wait

Q57. At a particular time, the value of a counting semaphore is 10, it will become 7 after:

- (a) 3 V operations
- (b) 3 P operations
- (c) 5 V operations and 2 P operations
- (d) 2 V operations and 5 P operations

Which of the following option is

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correct?

- (a) Only (b)
- (b) Only(d)
- (c) Both (b) and (d)
- (d) None of these

Q58. With single resource, deadlock occurs

- (a) If there are more than two processes competing for that resources
- (b) If there are only two processes competing for that resources
- (c) If there is a single process competing for that resources
- (d) None of these

Q59. At a particular time of computation the value of a counting semaphore is 7. Then 20 P operations and xV operations were completed on this semaphore. If the new value of semaphore is 5 ,x will be

- (a) 18
- (b) 22
- (c) 15
- (d) 13

Q60. A total of 9 units of a resource type available, and given the safe state shown below, which of the following sequence will be a safe state?

Process	Used	Max
P1	2	7
P2	1	6
P3	2	5
P4	1	4

- (a) (P4, P1, P3, P2)
- (b) (P4, P2, P1, P3)
- (c) P4, P2, P3, P1)
- (d) (P3, P1, P2, P4)

MODULE 4

Q61. Which of the following are the likely causes of thrashing?

- (a) Page size was very small
- (b) There are too many programs running in the system
- (c) Least recently used policy is used for page replacement
- (d) First in First out policy is used for page replacement

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Q62. In a 64-bit machine, with 2 GB RAM, and 8 KB page size, how many entries will be there in the page table if it is inverted?

- (a) 2^{18}
- (b) 2^{20}
- (c) 2^{33}
- (d) 2^{51}

Q63. Consider the list of page references in the timeline as below:

9 6 2 3 4 4 4 4 3 4 4 2 5 8 6 8 5 5 3 2 3 3 9 6 2 7

What is the working set at the penultimate page reference if Δ is 5?

- (a) { 8,5,3,2,9,6}
- (b) { 4,3,6,2,5}
- (c) { 3,9,6,2,7}
- (d) { 3,9,6,2,7}

Q64. Consider the following segment table in the segmentation scheme:

SegmentID	Base	Limit
0	200	200
1	500	1210
2	1527	498
3	2500	50

What happens if the logical address requested is Segment ID 2 and offset 1000?

- (a) Fetches the entry at the physical address 2527 for segment Id2
- (b) A trap is generated
- (c) Deadlock
- (d) Fetches the entry at offset 27 in Segment Id 3

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Q65. A computer has 16 pages of virtual address space but the size of main memory is only four frames. Initially the memory is empty. A program references the virtual pages in the order 0, 2, 4, 5, 2, 4, 3, 11, 2, 10. How many page faults occur if LRU page replacement algorithm is used?

- (a) 3
- (b) 5
- (c) 7
- (d) 8

Q66. Dirty bit is used to indicate which of the following?

- (a) A page fault has occurred
- (b) A page has corrupted data
- (c) A page has been modified after being loaded into cache
- (d) An illegal access of page

Q67. In a paged memory, the page hit ratio is 0.40. The time required to access a page in secondary memory is equal to 120 ns. The time required to access a page in primary memory is 15 ns. The average time required to access a page is

- (a) 105
- (b) 68
- (c) 75
- (d) 78

Q68. Which of the following statements are true ?

1. External Fragmentation exists when there is enough total memory space to satisfy a request but the available space is contiguous.
2. Memory Fragmentation can be internal as well as external.
3. One solution to external Fragmentation is compaction.

- (a) (a) and (b) only
- (b) (a) and (c) only
- (c) (b) and (c) only
- (d) (a), (b) and (c)

Q69. Page information in memory is also called as Page Table. The essential contents in each entry of a page table is/are

- (a) Page Access information
- (b) Virtual Page number
- (c) Page Frame number
- (d) Both virtual page number and Page Frame Number

Q70. Consider a virtual page reference string 1, 2, 3, 2, 4, 2, 5, 2, 3, 4. Suppose LRU page replacement algorithm is implemented with 3 page frames in main memory. Then the number of page faults are

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- (a) 5
- (b) 7
- (c) 9
- (d) 10

Q71. A CPU generates 32 bit virtual addresses. The page size is 4KB. The processor has a Translation Lookaside Buffer(TLB) which can hold a total of 128 page table entries and is 4-way set associative. The minimum size of the TLB tag is

- (a) 11 bits
- (b) 13 bits
- (c) 15 bits
- (d) 20 bits

Q72. Dirty bit is used to show the

- (a) Page with low frequency occurrence
- (b) Wrong page
- (c) Page with corrupted data
- (d) Page that is modified after being loaded into cache memory

Q73. If there are 32 segments, each size 1K bytes, then the logical address should have

- (a) 13 bits
- (b) 14 bits
- (c) 15 bits
- (d) 16 bits

Q74. Increasing the RAM of a computer typically improves performance because:

- (a) Virtual Memory increases
- (b) Larger RAMs are faster
- (c) Fewer page faults occur
- (d) Fewer segmentation faults occur

Q75. How many page faults occur in LRU page replacement algorithm for the given reference string, with four page frames

7,0,1,2,0,3,4,2,3,0,3,2,1,2,0,1

- (a) 6
- (b) 8
- (c) 7
- (d) 9

Q76. In a computer system, memory mapped access takes 100 nanoseconds when a page is found in TLB. In case the page is not TLB, it takes 400 nanoseconds to

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access. Assuming a hit ratio of 80

- (a) 120ns
- (b) 160ns
- (c) 200ns
- (d) 500ns

Q77.If a processor has 32 bit virtual address, 28 bit physical address, 2 KB pages. How many bits are required for the virtual,physical page number?

- (a) 17,21
- (b) 21,17
- (c) 6,10
- (d) None

Q78.The address sequence generated by tracing a particular program executing in a pure demand paging system with 100 records per page, with 1 free main memory frame is recorded as follows. 0100, 0200, 0430, 0499, 0510, 0530, 0560, 0120, 0220, 0240, 0260, 0320, 0370, What is the number of page faults?

- (a) 15,4
- (b) 6,4
- (c) 7,2
- (d) 4,6

Q79.Each process is contained in a single section of memory that is contiguous to the section containing the next process is called

- (a) Contiguous memory protection
- (b) Contiguous path name
- (c) Definite path name
- (d) Indefinite path name

Q80.Copying a process from memory to disk to allow space for other processes is called.....

- (a) Demand paging
- (b) Deadlock
- (c) page fault
- (d) Swapping

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Q81. In disk scheduling algorithm, the disk head moves from one end to other end of the disk, serving the requests along the way. When the head reaches the other end, it immediately returns to the beginning of the disk without serving any requests on the return trip.

- (a) LOOK
- (b) SCAN
- (c) C-LOOK
- (d) C-SCAN

Q82. An application programmer prefers using application programming interfaces rather than invoking system calls. Which of the following is NOT the correct reason for his preference?

- (a) APIs execute faster than system calls
- (b) APIs are easy to use
- (c) Use of APIs makes the program portable
- (d) APIs can potentially improve readability of the code

Q83. If the Disk head is located initially at track 32, find the number of disk moves required with FCFS scheduling criteria if the disk queue of I/O blocks requests are: 98, 37, 14, 124, 65, 67

- (a) 320
- (b) 322
- (c) 321
- (e) 319

Q84. Consider a disk with 16384 bytes per track having a rotation time of 16 msec and average seek time of 40 msec. What is the time in msec to read a block of 1024 bytes from this disk?

- (a) 57 sec
- (b) 49 sec
- (c) 48 sec
- (d) 17 sec

Q85. A disk drive has 100 cylinders, numbered 0 to 99. Disk requests come to the disk driver for cylinders 12, 26, 24, 4, 42, 8 and 50 in that order. The driver is currently serving a request at cylinder 24. A seek takes 6 msec per cylinder moved. How much seek time is needed for shortest seek time first (SSTF) algorithm?

- (a) 0.984 sec
- (b) 0.396 sec
- (c) 0.738 sec
- (d) 0.42 sec

Q86. Match the following

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- | | |
|--------------------------|-----------------|
| (a) Disk scheduling | (1) Round robin |
| (b) Batch processing | (2) Scan |
| (c) Time sharing | (3) LIFO |
| (d) Interrupt processing | (4) FIFO |

- (a) a-3, b-4, c-2, d-1
- (b) a-4, b-3, c-2, d-1
- (c) a-2, b-4, c-1, d-3
- (d) a-3, b-4, c-1, d-2

Q87. Consider a disk pack with 32 surfaces, 64 tracks and 512 sectors per pack. 256 bytes of data are stored in a bit serial manner in a sector. The number of bits required to specify a particular sector in the disk is

- (a) 19
- (b) 20
- (c) 18
- (d) 22

Q88. Assume that an implementation of Unix operating system uses i-nodes to keep track of data blocks allocated to a file. It supports 12 direct block addresses, one indirect block address and one double indirect block address. The file system has 256 bytes block size and 2 bytes for disk block address. The maximum possible size of a file in this system is

- (a) 16 MB
- (b) 16 KB
- (c) 70 KB
- (d) 71 KB
- (e) None of the Above

Q89. If the disk head is located initially at 32, find the number of disk moves required with FCFS if the disk queue of I/O blocks requests are 98, 37, 14, 124, 65, 67.

- (a) 239
- (b) 310
- (c) 321
- (d) 325

Q90. How much space will be required to store the bitmap of a 1.3 GB disk with 512 bytes block size ?

- (a) 332.8 KB
- (b) 83.6 KB

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(c)266.2 KB

(d)256.6 KB

Q91. Which is the primary purpose of disk structure management?

- (a) To increase file access time
- (b) To reduce fragmentation
- (c) To allocate contiguous blocks
- (d) To manage multiple threads

Q92. Which disk scheduling algorithm minimizes seek time by serving requests in one direction until the end?

- (a) FCFS
- (b) SSTF
- (c) SCAN
- (d) Circular SCAN

Q93. Which file system operation involves updating metadata like last modified time?

- (a) Create
- (b) Write
- (c) Open
- (d) Seek

Q94. Disk formatting prepares:

- (a) A disk for storage use
- (b) Pages for swapping
- (c) Memory for paging
- (d) Directories for execution

Q95. SSDs differ from magnetic disks by:

- (a) Having moving parts
- (b) Using flash memory
- (c) Using rotational storage
- (d) Being slower

Q96. File-system protection ensures:

- (a) File confidentiality
- (b) File integrity
- (c) Access control

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(d)All of the above

Q97. Directory implementation involves:

- (a)File name mapping
- (b) Memory segmentation
- (c)Deadlock prevention
- (d)Paging

Q98.Which is NOT a file operation?

- (a)Read
- (b) Write
- (c)Execute
- (d) Memory allocation

Q99.Disk scheduling aims to:

- (a) Minimize seek time
- (b)Format disks
- (c)Allocate memory
- (d)Defragment disks

Q100.The head of a hard disk serves requests following the shortest seek time first (SSTF) policy.

The head is initially positioned at truck number 180. Which of the request sets will cause the head to change its direction after servicing every request assuming that the head does not change direction if there is a tie in SSTF and all the requests arrive before the servicing starts?

- (a) 11, 139, 170, 178, 181, 184, 201, 265
- (b) 10, 138, 170, 178, 181, 185, 201, 265
- (c) 10, 139, 169, 178, 181, 184, 201, 265
- (d) 10, 138, 170, 178, 181, 185, 200, 265

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Answer Key

1. C	21. D	41. A	61. B	81. D
2. B	22. B	42. B	62. B	82. A
3. B	23. C	43. A	63. C	83. C
4. A	24. B	44. C	64. B	84. C
5. B	25. D	45. B	65. C	85. C
6. C	26. A	46. C	66. C	86. A
7. B	27. A	47. D	67. C	87. B
8. B	28. B	78. C	68. D	88. D
9. B	29. D	49. B	69. D	89. C
10. A	30. B	50. B	70. B	90. A
11. B	31. D	51. D	71. B	91. B
12. B	32. C	52. B	72. D	92. C
13. C	33. A	53. C	73. C	93. C
14. D	34. B	54. A	74. C	94. A
15. B	35. C	55. B	75. D	95. B
16. B	36. D	56. B	76. B	96. D
17. C	37. B	57. C	77. B	97. A
18. B	38. A	58. A	78. C	98. D
19. A	39. A	59. B	79. A	99. A
20. A	40. D	60. C	80. D	100. C